* Defn A set A is called a proper subset of set B if $A \subset B$ and $A \neq B$. We write $A \subsetneq B$

* Defn $\mathbb{N}$: The set of natural numbers
$$\mathbb{N} := \{1, 2, 3, \ldots\}$$

* Defn Set theoretic difference of A and B / The complement of B relative to A.
$$A \setminus B := \{x : x \in A \text{ and } x \notin B\}$$

* Defn Sets A and B are called disjoint if
$$A \cap B = \emptyset$$

* Lemma
$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$
$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$$

* Theorem (Principle of Induction). Let $P(n)$ be a statement depending on the natural number n. Suppose that

1. $P(1)$ is true
2. If $P(n)$ is true then $P(n+1)$ is true

Then $P(n)$ is true for all $n \in \mathbb{N}$

Hint: Use well ordering property of $\mathbb{N}$

* Defn Well ordering property of $\mathbb{N}$. Every non empty set of $\mathbb{N}$ has a smallest element.

* Defn Let A and B be sets. The Cartesian product is the set of tuples defined by
$$A \times B := \{(x, y) : x \in A, y \in B\}$$

* Defn A function $f : \underbrace{A}_{\text{domain of } f} \to \underbrace{B}_{\text{codomain of } f}$ is the subset f of $A \times B$ s.t for each $x \in A$ $\exists$ a unique $y \in B$ for which $(x,y) \in f$. We write $f(x) = y$

$$\underbrace{R(f)}_{\text{range of } f} := \{ y \in B : \exists x \in A \text{ s.t } f(x) = y \}$$

* Defn $f : A \to B$. Image / Direct Image of $C \subset A$

$$f(C) := \{ f(x) \in B : x \in C \}$$

Inverse Image of $D \subset B$

$$f^{-1}(D) := \{ x \in A : f(x) \in D \}$$

* Lemma $f : A \to B$. Let C, D be subsets of B then

$$f^{-1}(C \cup D) = f^{-1}(C) \cup f^{-1}(D)$$
$$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$$
$$f^{-1}(B \setminus C) = A \setminus f^{-1}(C)$$

* Lemma $f : A \to B$. Let C, D be subsets of A. then

$$f(C \cup D) = f(C) \cup f(D)$$
$$f(C \cap D) \subset f(C) \cap f(D)$$

* Defn Given a set A, a binary relation on A is a subset $R \subset A \times A$ which consists of those pairs where the relation holds. Instead of $(a,b) \in R$ we write aRb.

* Defn Let A and B be sets. We say A and B have the same cardinality when there exists a bijection $f : A \to B$.

* Defn If A have same cardinality as $\{1, 2, 3 \dots n\}$ for some $n \in \mathbb{N}$, we write $\underbrace{|A|}_{\text{equivalence class of all sets with same cardinality as } A} := n$. If A is empty we write $|A| := 0$] finite cardinality

* Defn We write $|A| \leq |B|$ if there exists an injection from A to B

* Defn If $|A| = \mathbb{N}$ we say A is countably infinite.
* Defn If A is finite or countably infinite we say A is countable

* Defn The power set of A, denoted by $P(A)$, is the set of all subsets of A.

* Theorem Let A be a set. Then $|A| < |P(A)|$. In particular there exists no surjection from A onto $P(A)$.

Proof Hint: $S = \{x \in A : x \notin f(x)\}$

Proof by contradiction: Assume a surjective map exists

Let $c \in A$ s.t $f(c) = S$

Defn :- An ordered set is a set S together with a relation '$<$' s.t

① $\forall x, y \in S$, exactly one of the following holds, ie.

$x<y$ or $x>y$ or $x=y$

② If $x, y, z \in S$ s.t $x<y$ and $y<z$ then $x<z$

Defn Let $E \subset S$ where S is an ordered set

* If $\exists b \in S$ s.t $\forall x \in E$ $x \le b$, b is called an upper bound of E
* If $\exists b_0 \in S$ s.t b_0 is an upper bound of E and if b is another upper bound of E then $b_0 \le b$ then b_0 is called least upper bound or supremum of E
* If $\exists b \in S$ s.t $\forall x \in E$ $x \ge b$, then b is called an lower bound of E
* If $\exists b_0 \in S$ s.t b_0 is an lower bound of E and if b is another lower bound of E then $b \le b_0$, then b_0 is called greatest lower bound or infimum of E

Defn An ordered set S has the least upper bound property if every non empty set $E \subset S$ that is bounded above has a least upper bound

Defn A set F is called field if it has two operations, addition $(+)$ and multiplication, defined on it s.t

① If $x, y \in F$ then $x+y \in F$
② $\forall x, y \in F$ $x+y = y+x$
③ $\forall x, y, z \in F$ $(x+y)+z = x+(y+z)$
④ $\exists\ 0 \in F$ s.t $\forall x \in F$ $x+0 = 0+x = x$
⑤ $\forall x \in F$ $\exists -x \in F$ s.t $x+(-x) = 0$

Abelian group under addition

⑥ If $x, y \in F$ then $x.y \in F$
⑦ $\forall x, y \in F$ $x.y = y.x$
⑧ $\forall x, y, z \in F$ $(x.y).z = x.(y.z)$
⑨ $\exists\ 1 \in F$ s.t $1 \ne 0$ and $\forall x \in F$ $1.x = x.1 = x$
⑩ $\forall x \in F$ s.t $x \ne 0$ $\exists x^{-1} \in F$ s.t $x.x^{-1} = x^{-1}.x = 1$

Abelian group under multiplication

⑪ $\forall x, y, z \in F$ $x.(y+z) = xy + xz$

* **Defn** An field F is called an ordered field if it is an ordered set s.t

① $\forall\, x, y, z \in F \quad y<z \Rightarrow x+y < x+z$

② If $x, y \in F$ s.t $x>0$ and $y>0$ then $xy>0$

* **Lemma** Let F be an ordered field and $x, y, z, w \in F$. Then

① If $x>0$ and $y<z$ then $xy<xz$

② If $x \neq 0$ then $x^2>0$ corollary $1>0$ Proof Hint $-x=-1x \rightarrow -1x-1=1$

③ If $0<x<y$ then $0<\frac{1}{y}<\frac{1}{x}$

④ $0<x<y$ then $x^2<y^2$

⑤ If $x \leq y$ and $z \leq w$ then $x+z \leq y+w$

* **Lemma** Let F be an ordered field with least upper bound property. If $A \subset F$ s.t A is non empty and bounded below then $\inf(A)$ exists.

* **Defn** There exists a unique ordered field $\mathbb{R}$ with least upper bound property s.t $\mathbb{Q} \subset \mathbb{R}$

* **Lemma** If $x \in \mathbb{R}$ s.t $x \leq \varepsilon$ $\forall \varepsilon \in \mathbb{R}$ where $\varepsilon>0$ then $x \leq 0$

* **Theorem** If $x, y \in \mathbb{R}$ and $x>0$ then $\exists\, n \in \mathbb{N}$ s.t $nx>y$
(Archimedian Property)

Proof Hint: Rephrase $\mathbb{N} \subset \mathbb{R}$ doesn't have an upper bound

contradiction: Let $b := \sup(\mathbb{N})$ $\exists\, m > b-1$ where $m \in \mathbb{N}$

$m+1 > b$

* **Theorem** If $x, y \in \mathbb{R}$ and $x<y$ then $\exists$ an $r \in \mathbb{Q}$ s.t $x<r<y$
($\mathbb{Q}$ is dense in $\mathbb{R}$)

Proof Hint Let $x>0$ the $\exists n \in \mathbb{N}$ s.t $n(y-x)>1$ —①

Define $A := \{k \in \mathbb{N} : k > nx\}$. $A \subset \mathbb{N}$ well ordering prop.

$m \in A$ is the smallest element $m > nx \Rightarrow x < m/n$

$m-1 \leq nx \rightarrow m \leq nx+1 \underbrace{<}_{\text{from 1}} ny \Rightarrow m/n < y$

* **Defn** Let $A \subset \mathbb{R}$ and $x \in \mathbb{R}$

$$x + A := \{ x+y \in \mathbb{R} : y \in A \}$$
$$xA := \{ xy \in \mathbb{R} : y \in A \}$$

* **Lemma** If A is bounded above $\sup(x+A) = x + \sup(A)$

" " " " below $\inf(x+A) = x + \inf(A)$

If $x > 0$, A is bounded above $\sup(xA) = x \sup(A)$

" $x > 0$, A " " below $\inf(xA) = x \inf(A)$

" $x < 0$, A " " above $\inf(xA) = x \sup(A)$

" $x < 0$, A " " below $\sup(xA) = x \inf(A)$

* **Lemma** Let $A, B \subset \mathbb{R}$ be non empty subsets s.t $x \leq y$ $\forall x \in A$ and $y \in B$, then

$$\sup(A) \leq \inf(B)$$

* **Lemma** If $S \subset \mathbb{R}$ is a non empty set, bounded above, then $\forall \varepsilon > 0$ $\exists$ $x \in S$ s.t

$$\sup(S) - \varepsilon < x \leq \sup(S)$$

* **Defn** Let $A \subset \mathbb{R}$ be a set

(i) If A is empty then $\sup(A) := -\infty$

$\inf(A) := \infty$

(ii) If A is not bounded above then $\sup(A) := \infty$

(iii) " " " " " below then $\inf(A) := -\infty$

* **Defn** When a set $A \subset \mathbb{R}$ is bounded above s.t $\sup(A) \in A$, then we denote $\sup(A)$ by $\max(A)$. Similarly $\min(A)$ is defined

* **Defn** Absolute value function $|x| := \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

* **Lemma**
  * $|x| \geq 0$ and $|x| = 0$ if and only if $x = 0$
  * $|xy| = |x||y|$
  * $|x| \leq y$ if and only if $-y \leq x \leq y$
  * $-|x| \leq x \leq |x|$ $\forall x \in \mathbb{R}$

* **Lemma**
  * $|x+y| \leq |x| + |y|$ $\forall x, y \in \mathbb{R}$
  * $|x-y| \leq |x| + |y|$ "
  * $||x| - |y|| \leq |x-y|$ "

* **Defn** $f : D \to \mathbb{R}$. We say f is bounded if $\exists$ $M \in \mathbb{R}$ st

$$|f(x)| \leq M \qquad \forall x \in D$$

* **Lemma** If $f : D \to \mathbb{R}$ and $g : D \to \mathbb{R}$ are bounded functions and $f(x) \leq g(x)$ $\forall x \in D$, then

$$\sup_{x \in D} f(x) \leq \sup_{y \in D} g(y) \qquad \inf_{x \in D} f(x) \leq \inf_{y \in D} g(y)$$

* **Defn** Intervals

$$[a,b] = \{x \in \mathbb{R} \text{ st } a \leq x \leq b\}$$

$$(a,b] = \{x \in \mathbb{R} \text{ st } a < x \leq b\}$$

* **Defn** All the intervals have the same cardinality

Example $\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to (-\infty, \infty)$ is a bijective map

$\therefore$ $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ have same cardinality as $\mathbb{R}$

* Theorem $\mathbb{R}$ is uncountable

Proof Hint :- Let $X \subset \mathbb{R}$ be countably infinite s.t if $a<b$ and $a, b \in \mathbb{R}$ then $\exists x \in X$ s.t $a<x<b$

countably infinite $\therefore \exists$ a bijection $f: \mathbb{N} \to X$ Let $x_1 = f(1)$

$a_1 := x_1$ $b_1 := x_1 + 1$

Construct $a_1 < a_2 < a_3 \dots < a_n < b_n < b_{n-1} \dots < b_2 < b_1$

s.t $x_j \notin (a_k, b_k)$ where $j = 1, 2, \dots, k$

$A = \{a_1, a_2 \dots\}$ $B = \{b_1, b_2 \dots\}$

$y = \sup(A) \le \inf(B)$

Prove $y \notin A$ and $y \notin B$ and $y \notin X$

$\therefore X$ cannot contain all elements of $\mathbb{R}$

$\therefore \mathbb{R}$ is uncountable

* Defn For infinite sequence of digits $0.d_1 d_2 d_3 \dots$

$$D_n = \frac{d_1}{10} + \frac{d_2}{10^2} \dots + \frac{d_n}{10^n}$$

Lemma * Every infinite sequence of digits $0.d_1 d_2 \dots$ represents a unique real number $x \in [0, 1]$, s.t

$$D_n \le x \le D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

** For every $x \in (0, 1]$ $\exists$ a unique infinite sequence of digits $0.d_1 d_2 d_3 \dots$ that represents x s.t

$$D_n < x \le D_n + \frac{1}{10^n} \quad \forall n \in \mathbb{N}$$

* Defn A sequence of real numbers is a function $x: \mathbb{N} \longrightarrow \mathbb{R}$.

Another notation $\{x_n\}_{n=0}^{\infty}$

* Defn A sequence $\{x_n\}_{n=1}^{\infty}$ is called bounded if the underlying function is bounded.

* Defn A sequence $\{x_n\}_{n=0}^{\infty}$ is said to converge to a $x \in \mathbb{R}$ if $\forall \varepsilon > 0, \varepsilon \in \mathbb{R}$ $\exists M \in \mathbb{N}$ s.t $|x_n - x| < \varepsilon$ $\forall n \geq M$.

Limit of the sequence

$$\lim_{n \to \infty} x_n := x$$

* Defn A sequence that is not convergent is called divergent

* Lemma A convergent sequence has a unique limit.

* Lemma A convergent sequence is bounded

Proof Hint :- Use triangle inequality

* Defn $\{x_n\}_{n=0}^{\infty}$ is montone increasing if $x_{n+1} \geq x_n$ $\forall n \in \mathbb{N}$

" " " decreasing if $x_{n+1} \leq x_n$ $\forall n \in \mathbb{N}$

* Theorem A monotone sequence is bounded if and only if it is convergent

if $\{x_n\}_{n=1}^{\infty}$ is monotone increasing $\lim_{n \to \infty} x_n = \sup\{x_n : n \in \mathbb{N}\}$

" " " " decreasing " " $= \inf\{x_n : n \in \mathbb{N}\}$

* Lemma Let $S \subset \mathbb{R}$ be a nonempty bounded set. Then $\exists$ monotone sequences $\{x_n\}_{n=0}^{\infty}, \{y_n\}_{n=0}^{\infty}$ s.t $x_n, y_n \in S$ and

$$\sup(S) = \lim_{n \to \infty} x_n$$

$$\inf(S) = \lim_{n \to \infty} y_n$$

* Def$_n$: For a sequence $\{x_n\}_{n=1}^{\infty}$, the K-tail, where $K \in \mathbb{N}$, is the sequence
$$\{x_{n+K}\}_{n=1}^{\infty} \quad \text{or} \quad \{x_n\}_{n=K+1}^{\infty}$$

* Lemma For $\{x_n\}_{n=1}^{\infty}$ the following statements are equivalent;
  * $\{x_n\}_{n=1}^{\infty}$ converges
  * The K tail $\{x_{n+K}\}_{n=1}^{\infty}$ converges for $\forall K \in \mathbb{N}$
  * " " " " converges for some $K \in \mathbb{N}$

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} x_{n+K}$$

* Def$_n$ Let $\{x_n\}_{n=1}^{\infty}$ be a sequence. Let $\{n_i\}_{i=1}^{\infty}$ be a strictly increasing sequence of natural number.

$$\{x_{n_i}\}_{i=1}^{\infty} \quad \text{is called a subsequence}$$

* Lemma If $\{x_n\}_{n=1}^{\infty}$ is a convergent sequence. Then every subsequence $\{x_{n_i}\}_{i=1}^{\infty}$ is convergent.

* Lemma Let $\{a_n\}_{n=1}^{\infty}$, $\{b_n\}_{n=1}^{\infty}$ and $\{x_n\}_{n=1}^{\infty}$ be sequences s.t
  1. $\{a_n\}_{n=1}^{\infty}$ and $\{b_n\}_{n=1}^{\infty}$ are convergent
  2. $\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$
  3. $a_n \leq x_n \leq b_n \quad \forall n \in \mathbb{N}$

then

$$\{x_n\}_{n=1}^{\infty} \text{ is convergent and } \lim_{n\to\infty} x_n = \lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n$$

* Lemma Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ be convergent series s.t $x_n \leq y_n$ $\forall n \in \mathbb{N}$ then

$$\lim_{n\to\infty} x_n \leq \lim_{n\to\infty} y_n$$

* **Lemma** Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ be convergent sequences

(i) The sequence $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n + y_n$ is convergent and

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$$

(ii) The sequence $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n - y_n$ is convergent

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n - \lim_{n\to\infty} y_n$$

(iii) The sequence $\{z_n\}_{n=1}^{\infty}$ where $z_n := x_n y_n$ is convergent

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty} x_n \cdot \lim_{n\to\infty} y_n$$

(iv) If $\lim_{n\to\infty} y_n \neq 0$ and $y_n \neq 0 \ \forall n \in \mathbb{N}$, then $\{z_n\}_{n=1}^{\infty}$ where $z_n = \frac{x_n}{y_n}$ is convergent and

$$\lim_{n\to\infty} z_n = \frac{\lim_{n\to\infty} x_n}{\lim_{n\to\infty} y_n}$$

* **Lemma** If $\{x_n\}_{n=1}^{\infty}$ is convergent sequence s.t $x_n \geq 0 \ \forall n \in \mathbb{N}$ then

$$\lim_{n\to\infty} \sqrt{x_n} = \sqrt{\lim_{n\to\infty} x_n}$$

* **Lemma** If $\{x_n\}_{n=1}^{\infty}$ is convergent sequence then $\{|x_n|\}_{n=1}^{\infty}$ is also convergent, further

$$\lim_{n\to\infty} |x_n| = |\lim_{n\to\infty} x_n|$$

* **Lemma** Let $\{x_n\}_{n=1}^{\infty}$ be a sequence. Suppose $\exists x \in \mathbb{R}$ and a convergent sequence $\{a_n\}_{n=1}^{\infty}$ s.t

① $\lim_{n\to\infty} a_n = 0$

② $|x_n - x| < a_n \quad \forall n \in \mathbb{N}$

Then $\{x_n\}_{n=1}^{\infty}$ is convergent and $\lim_{n\to\infty} x_n = x$

* Lemma   Let $c > 0$

(i) If $c < 1$ then the sequence $\{c^n\}_{n=1}^{\infty}$ converges to 0

(ii) If $c > 1$ then the sequence $\{c^n\}_{n=1}^{\infty}$ is unbounded

* Lemma Let $\{x_n\}_{n=1}^{\infty}$ be a sequence st $x_n \neq 0$ $\forall n \in \mathbb{N}$ and the following limit exists

$$\lim_{n \to \infty} \frac{|x_{n+1}|}{|x_n|}$$

(i) If $L < 1$ then $\{x_n\}_{n=1}^{\infty}$ is convergent $\lim_{n \to \infty} x_n = 0$

(ii) If $L > 1$ then $\{x_n\}_{n=1}^{\infty}$ is unbounded